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Beyond Detection in Wastewater-Based Epidemiology: Sewer Processes, Signal Distortion, Normalization, and Epidemiological Inference

1. Introduction and review approach

1.1 The measurement-inference gap

Wastewater-based epidemiology (WBE) rests on a premise that has proved harder to operationalize than its early, detection-oriented literature assumed: a concentration measured at a treatment-plant inlet is not itself a public health indicator. It becomes one only after a chain of transformations — human excretion, building drainage, sewer transport, sampling, and analysis — has been accounted for and, where possible, inverted. Recent methodological work converges on this point from several directions at once: population-normalization markers show no consensus best choice (Boogaerts et al., 2024), flow-rate normalization outperforms common fecal biomarkers in some large comparisons but not others (Rainey et al., 2023), and in-sewer microbial activity is now understood to drive substantial, distance-dependent viral ribonucleic acid (RNA) loss during transport (Jung et al., 2026). Read together, this literature marks a shift in the field’s central question, from *what can be detected in wastewater* to *under what conditions does a measured concentration reliably represent the upstream state it is assumed to track*.

This review treats that question explicitly as an inverse problem. A wastewater sample is an observation generated by an unknown source signal passed through a chain of physical operators, each capable of distorting the signal in one or more of three formal ways used throughout this review. **Amplitude distortion** covers multiplicative or additive changes to signal magnitude — dilution, decay, recovery loss, matrix effects. **Temporal distortion** covers shifts, lags, or smearing of the signal in time — variable shedding kinetics, sewer retention-time distributions, sampling-schedule mismatch. **Spatial distortion** covers changes in which sub-populations and sub-catchments contribute, and in what proportion, to a single downstream composite sample — heterogeneous flow paths, combined-sewer overflow, differential storm response across a network. Recovering the source signal from a distorted, aggregated observation is possible only to the extent the problem remains **identifiable** — that is, to the extent the available data constrain the space of source signals consistent with what was actually measured. Where identifiability is poor, **process-informed correction** — using mechanistic models of excretion, transport, and decay, rather than purely statistical normalization, to narrow that space — is this review’s proposed response, though it cannot guarantee identifiability or recover information already lost during transport, sampling, or aggregation.

1.2 Scope and critical-review approach

This is a narrative synthesis organized around the mechanistic framework above, not yet a formal systematic or scoping review; a defensible scoping review of this literature would follow PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews), use structured Boolean database queries, apply documented inclusion and exclusion criteria, and report a PRISMA flow diagram, none of which has been performed at this stage. Evidence was instead identified through targeted web search, screened by title and snippet, prioritizing peer-reviewed journal articles and institutional publications over secondary sources. This approach establishes a defensible starting evidence set and confirms the review’s mechanistic framing has real literature behind it, but almost certainly under-samples the field, particularly older foundational work and non-English-language studies. The full search strategy and source-verification

procedure are given in Supplementary Methods; every specific number below should be read against the limitations stated there rather than as independently confirmed.

Rather than cataloging every WBE target analyte or disease application, this review is scoped to the signal-formation, distortion, and correction problem shared across targets — pathogens, antimicrobial-resistance markers, and chemical or consumption biomarkers alike — using specific examples to illustrate a shared structural argument, not as the organizing structure itself.

1.3 Conceptual contributions

This review advances four linked propositions, offered as this review’s own organizing synthesis rather than as an existing consensus. First, **WBE is usefully formalized as an inverse problem**, in which a source-end signal is recoverable from a downstream measurement only to the extent the intervening transformation chain is identifiable. Second, every distortion mechanism in that chain — source-end, sewer-transport, sampling, and analytical — can be organized under one consistent **amplitude-temporal-spatial taxonomy**, rather than treated as a disconnected list of stage-specific error sources. Third, **normalization is best understood as a mechanism-specific causal correction, not a reconstruction of the source state**: a normalization step removes only the distortion it is mechanistically linked to, and treating it as a general-purpose fix is a documented, recurring source of error in the applied literature. Fourth, reconstruction and the decisions built on it benefit from an **identifiability- and uncertainty-aware framework**, distinguishing association, predictive performance, parameter identifiability, state observability, and structural adequacy as five separate kinds of evidence a reconstruction claim can be asked to meet, carried forward into how a reconstructed, uncertainty-quantified signal should inform a public-health decision. Taken together, these four propositions target a single recurring failure mode this review identifies in the applied literature it surveys: a result established at one stage of the source-to-decision chain — a detection, a correlation, a narrow confidence interval — being read as though it settled a question that in fact belongs to a later, evidentially distinct stage.

2. From population states to wastewater signals

Population-level WBE inference begins from an underlying state in the contributing population — **infection status, consumption or exposure**, or a broader health-status indicator — that is true but unmeasured at any given time. WBE’s premise is that this state produces a mass of measurable analyte via **metabolism and excretion**, entering the sewer network in a form assumed to bear a stable, recoverable relationship to the state itself. That assumption is only approximately true, and where it breaks down determines which distortion the rest of this review must correct for.

Source-end variation is itself a source of temporal and amplitude distortion, not measurement noise to be averaged away. For infectious targets, individual shedding is neither universal nor uniform: concentration and duration vary substantially across infected individuals, with a subset of high-intensity shedders contributing disproportionately to the aggregate signal — a right-skewed, over-dispersed pattern, not one well summarized by a mean and a standard deviation. Because the population mix of early-, peak-, and late-stage infections shifts over the course of an outbreak, the same number of currently-infected individuals can correspond to different total excreted mass on different days: this broadens the effective source pulse relative to the sharper underlying incidence curve it is meant to track, a temporal distortion in the formal sense above. For chemical targets — pharmaceuticals, illicit drugs — excreted mass instead depends on administered dose, route, individual metabolic rate, and the fraction of parent compound excreted as the specific biomarker measured; because this

pharmacokinetic correction factor is typically derived from clinical studies in smaller, more homogeneous populations than the sewershed being monitored, it introduces a systematic amplitude bias the wastewater data alone cannot detect.

WBE's core comparative advantage is, at the same time, an identifiability complication. Because wastewater signal captures shedding regardless of symptom status or care-seeking behavior, it includes asymptomatic and undiagnosed infections that clinical surveillance systematically misses (NASEM, 2023). The two signals therefore do not share a denominator: treating wastewater and clinical case counts as measuring the same underlying quantity, when they structurally do not, is an identifiability error, not merely a calibration detail to be resolved with a scaling constant.

Excretion and shedding are properly represented as distributions, not point values, and the shape of that distribution matters as much as its mean. A population-average shedding rate used in a back-calculation model is not a description of any single individual; it is a summary of a mixture of individuals at different points on their own shedding trajectory at any given observation time, and treating that mixture as approximately normal — rather than the right-skewed, over-dispersed shape the underlying biology actually produces — understates exactly the tail behavior that determines outbreak-detection sensitivity at low prevalence. The same distributional caution applies to chemical targets: the ratio of parent compound to excreted metabolite varies with dose, co-administered substances, and individual metabolic-enzyme activity, layering a further, independent source of dispersion on top of the dose-and-route variation already noted above.

The contributing population is itself dynamic, not a fixed denominator. Commuting, tourism, institutional populations, and seasonal movement mean the population actually producing a given day's signal is rarely identical to a catchment's registered residents; how this fluctuation is estimated and corrected for is developed later in this review (§4), but the underlying phenomenon — a moving, not fixed, denominator — belongs at the source-state level, since it changes what a given excretion total represents before any correction is applied to it.

Between excretion and the public sewer lies building plumbing — the stage closest to the source, and in principle the most tractable at which to characterize distortion, yet one largely absent from the applied uncertainty literature. Per-fixtured flush volumes and building water-use efficiency set an initial dilution ratio before any excreta reaches the public network; institutional sources — hospitals, schools, elder-care facilities, prisons — have occupant excretion, medication-use, and disinfectant-use patterns that differ systematically from residential norms and can locally dominate a sub-catchment's signal; and finite in-building retention time allows limited transformation to begin before wastewater ever reaches the public sewer. Because building- and sub-catchment-scale monitoring is a rapidly growing application for institutional outbreak localization, building-level distortion is increasingly a monitoring-design question in its own right, not only a theoretical caveat.

Non-human and non-ingested sources are a further, source-adjacent distortion specific to chemical targets. A fraction of measured mass may originate from unused-medication disposal or veterinary use rather than human metabolism, behaving as a source-end bias term that no excretion-fraction correction, however well characterized, can detect or remove. The same category, for pathogen and antimicrobial-resistance targets, includes non-human animal or environmental contributions entering a network upstream of any human source — a confound that a purely concentration-based signal cannot separate from genuine human-source material without additional source-attribution information.

The signal actually entering the public sewer network — the sewer-entry load — is therefore already a distorted copy of the true source state, before any of the transport, sampling, or analytical processes the remainder of this review addresses have acted on it at all. Keeping this distinction explicit throughout — source state versus sewer-entry load — is what allows later sections to attribute a given distortion to the stage that actually produced it, rather than absorbing every discrepancy into a single undifferentiated measurement uncertainty.

3. Sewers as information filters

3.1 Hydraulic transport and residence-time distributions

Sewage in transit does not move as a single, uniform slug. Different tributary paths converging at one downstream sampling point carry different, and independently uncertain, hydraulic retention times, so a single treatment-plant sample is a time- and path-weighted composite whose weighting is rarely known explicitly. This residence-time distribution is not a fixed delay but itself a distribution over arrival times, varying with pipe diameter and slope, sediment accumulation, and pump-station or storage operation — meaning two sewersheds with identical source-end populations can produce measurably different treatment-plant signals from network geometry and operation alone. Characterizing this distribution, through hydraulic modeling or tracer studies, is a prerequisite for any correction that claims to be process-informed rather than purely statistical.

This matters beyond transport bookkeeping because the residence-time distribution is the formal object that determines how much a sharp, source-end signal is smeared before it is ever observed. A network with a wide, heterogeneous residence-time distribution across its contributing sub-catchments will blur a genuine, sharp epidemiological signal — an outbreak onset, a sudden consumption spike — into a slower, harder-to-localize rise, independent of anything happening at the source. Reporting a treatment-plant signal without any characterization of the residence-time distribution behind it is, in effect, reporting a temporally distorted signal without disclosing the distortion's likely magnitude.

3.2 Transformation, partitioning and biological interactions

Physical, chemical, and biological processes each reshape a target signal during transit, and they do not act independently of one another. Physically, dilution from base infiltration, particle adsorption and desorption governing liquid-solid partitioning, and incomplete mixing across converging sub-catchment flows all alter the signal before it is ever sampled. Chemically, hydrolysis, redox reactions, and pH- and ionic-strength-dependent speciation can change a chemical target's measurable concentration independent of any change in the mass originally excreted. Biologically — and this is the strongest single mechanistic finding in this review's evidence base — sewer biofilms are an active sink for viral RNA rather than an inert surface: laboratory reactor studies report substantial RNA loss within hours of biofilm exposure (Zhang et al., 2023), and combined batch-decay and pipeline-simulator work identifies microbial activity, rather than chemical hydrolysis, as the dominant driver of that loss, with decay increasing measurably as a function of transport distance (Jung et al., 2026). Because decay is distance-, and therefore retention-time-, dependent, amplitude loss is not spatially uniform across a network: sub-catchments with longer retention times lose proportionally more signal, a coupling that a purely multiplicative flow-normalization step cannot, by construction, correct.

The rate and extent of every one of these transformations is jointly governed by network-level operating conditions — wastewater temperature, dissolved oxygen and redox state, pH, hydraulic retention time, pipe diameter and slope (which set flow velocity and hence both retention time and shear-driven resuspension), sediment accumulation, and whether a network is combined or separated. None of these is a property of the target analyte; all are properties of the network the target happens to be traveling through, which is precisely why the same pathogen or chemical marker can show materially different in-transit behavior between two sewersheds, or even between two sub-catchments of the same sewershed, independent of any difference in the true source-end signal.

3.3 Sedimentation, resuspension and memory effects

A target adsorbed to sediment during low-flow conditions can later be resuspended during a subsequent flow event, producing a signal spike that reflects historical, not concurrent, source material. This is mechanistically distinct from ordinary dilution or decay: it gives the sewer network a form of memory, in which a downstream sample at a given time can reflect source-end conditions from well before that time. No same-day auxiliary variable — a same-day flow reading, a same-day population estimate — can distinguish a resuspension-driven spike from a genuine concurrent rise in the source signal, because both produce the same superficial signature (a rise in measured concentration) from mechanistically unrelated causes. Distinguishing them in practice requires either an independent line of evidence about the resuspension event itself (a flow or turbidity record consistent with a recent disturbance) or an explicit acknowledgment that the two explanations cannot be separated from the concentration record alone — treating a resuspension-driven spike as a confirmed source-signal rise, without that check, is a specific and avoidable interpretation error.

3.4 Rainfall, infiltration, overflow and warm-season forcing

Rainfall is not merely dilution. It enters sewer networks through two structurally different pathways — fast, direct inflow through connected drains, and slower groundwater infiltration through pipe joints — that together dilute measured concentration, but also resuspend settled sediment, shorten hydraulic retention time by increasing flow velocity, and shift the relative contribution of different sub-catchments to the composite signal, as normally minor flow paths become proportionally larger during storms. In combined sewer systems, sustained rainfall can trigger overflows that vent wastewater directly to the environment before it reaches the sampling point — a loss pathway that depresses both measured concentration and flow simultaneously, capping rather than merely diluting the true signal.

This compound perturbation is not, however, unstructured noise. A large multi-plant national wastewater surveillance program found that explicitly correcting concentration data for rainfall impact substantially improved correlation with independent clinical case counts relative to uncorrected data, and that uncorrected rainy-period data systematically underestimated the underlying signal (Janssens et al., 2022) — direct evidence that rainfall behaves as a correctable, structured bias, not a source of irreducible error. Elevated temperature acts through an overlapping but mechanistically distinct pathway, independently accelerating both target decay and sewer-biofilm metabolic activity (Guo et al., 2023; Jung et al., 2026, §3.2). Because rainfall and temperature are frequently seasonally correlated — warm-season convective storms in many climates — the two mechanisms often co-occur in practice and are difficult to separate empirically without an explicit mechanistic transport-and-decay model, itself an argument for process-informed rather than purely statistical correction.

3.5 Amplitude, temporal and spatial distortion

Sections 3.1–3.4 are not independent nuisance factors correctable one at a time with a single ratio: retention time affects decay, decay is spatially heterogeneous, and rainfall shifts retention time and sub-catchment weighting simultaneously. This coupling is precisely the condition under which an inverse problem becomes poorly identified — many different combinations of true source signal, decay history, and sub-catchment weighting are consistent with the same single observed time series, especially when that series is sampled at low frequency relative to how quickly these processes change.

Two conclusions follow, and both point the same direction. First, a single fixed correction ratio cannot restore identifiability under these conditions, because it implicitly assumes the amplitude and temporal distortion terms are constant, or vary only with flow, when the evidence above shows they vary with temperature, retention time, and storm state as well. This is consistent with, not contradicted by, flow normalization outperforming biomarker normalization in some large multi-site comparisons (Rainey et al., 2023) while becoming unreliable under substantial inflow and infiltration in others (Darling et al., 2025): the two findings together indicate that no single fixed correction factor is identifiable across all hydraulic conditions,

rather than that one method is generally superior to the other. Second, process-informed correction — explicitly modeling amplitude as a function of measured temperature, flow, and rainfall, and the transport kernel as a function of measured or estimated retention-time distribution — is what narrows the solution space back to something usable, as demonstrated in the one case in this evidence base where a mechanistic rainfall correction was directly tested against an uncorrected alternative (Janssens et al., 2022).

The sewer network, in short, is an active information filter, not a passive conduit. It does not merely delay and dilute a source signal on its way to the treatment plant; it reshapes which distortions are even in principle correctable by any downstream statistical step, which is the argument the rest of this review develops in quantitative terms.

4. Observation and normalization

4.1 Sampling and analytical observation

What reaches a treatment plant is sampled and analyzed before it is ever reported as a number, and both steps introduce their own, mechanistically distinct distortion. Sampling **mode** matters as much as sampling location: a grab sample captures a single instant and is vulnerable to diurnal and storm-driven swings in concentration, while a flow- or time-proportional composite sample averages over a collection window at the cost of temporal resolution — the two are not interchangeable, and which is appropriate depends on whether the inference goal is trend detection, which tolerates averaging, or anomaly detection, which needs finer temporal resolution. Sampling **timing** relative to diurnal and storm cycles determines what a single sample can represent, independent of downstream correction.

At the analytical stage, **recovery** — the fraction of true target mass that survives concentration and extraction — and **inhibition or matrix suppression** — interference with the detection chemistry itself — are mechanistically distinct amplitude-loss terms: a sample can have excellent recovery and still be strongly inhibited, or vice versa, which is why they require separate quality-control checks rather than one combined correction factor. These distinctions matter because they determine which of three different downstream consequences a given error produces. **Systematic bias** is a persistent, directional offset not reduced by averaging. **Random error** is sample-to-sample noise, reduced by replication. **Information loss** is a distortion severe enough — values censored below a detection limit, samples spatially pre-aggregated before collection — that no amount of correction recovers what was never captured in the first place. An explicit **observation model**, specifying which correction terms apply to a given target class (molecular versus chemical), is this review’s device for keeping these three consequences, and the corrections appropriate to each, from collapsing into one undifferentiated “measurement uncertainty.” Six concepts recur from this point forward and stay distinct throughout: the three consequences already introduced — **systematic bias, random error, information loss** — and three common upstream sources — **recovery loss, inhibition, and temporal or compositional representativeness** (whether a sample actually reflects the condition it is taken to represent). Part 5 propagates each through to a final uncertainty estimate.

4.2 Flow and population normalization

The most direct normalization step converts a measured concentration into a mass load using measured flow, correcting for dilution. Flow measurement is itself imperfect and location-dependent, and a flow-normalized load is never more precise than its noisier input; an instantaneous flow reading paired with a time-averaged composite concentration introduces a further temporal mismatch.

Flow normalization corrects dilution, not population size. Converting a load to a per-capita quantity additionally requires a population estimate, and the choice between a static census figure and a dynamic estimate is a choice about which distortion the correction can and cannot address. A sewershed's design, census, and actually-connected populations are three different numbers, and commuting, tourism, and institutional populations shift the population truly served relative to any of them. Census-based normalization silently reinterprets real population fluctuation as a change in per-capita source signal — precisely backwards for any application trying to detect a true epidemiological change during a population shift. Dynamic population estimation, via mobile-device data or a chemical proxy such as ammonium, corrects this but introduces the proxy's own measurement uncertainty as a new denominator-error term, and raises its own representativeness and privacy concern at small spatial scales, where a highly precise dynamic population estimate can approach re-identifying information about specific households.

Rainfall, infiltration, and industrial or non-domestic discharge each limit what flow normalization alone can fix, and each requires separate handling rather than being assumed away by a single flow term. Comparative evidence shows flow normalization outperforming biomarker alternatives in some large multi-site studies (Rainey et al., 2023) while becoming unreliable under substantial inflow and infiltration in others (Darling et al., 2025) — a context-dependent applicability condition, not a universal ranking.

4.3 Biomarker normalization and fate matching

Where a chemical or microbial population biomarker is used instead of, or alongside, flow, the correction is valid only to the extent the biomarker and the target share the distortion mechanism the normalization is meant to remove. No single population biomarker has been validated as universally reliable: the evidence base shows a consistent pattern of candidates being proposed, then found to fail specific validation criteria — human specificity, excretion stability, dietary or behavioral dependence, in-sewer persistence, and rainfall sensitivity are not correlated, so a marker can pass some criteria and fail others. Creatinine, for instance, is physiologically well-motivated but disqualified by in-sewer instability; cotinine tracks population reasonably well but is confounded by geographically variable smoking prevalence; Pepper Mild Mottle Virus (PMMoV) and crAssphage are widely used but show inconsistent, often neutral-or-negative, performance across independent multi-site validation studies.

The mechanistic requirement underlying all of this is **fate matching**: a target and its normalizer must share comparable persistence, solid-liquid partitioning, and temperature sensitivity for their ratio to track a real dilution or population signal, rather than a divergence in their own independent transport and decay behavior. Because a ratio's numerator and denominator each carry independent measurement error, and because a ratio's sensitivity to denominator error grows sharply as the denominator approaches its own detection limit, a normalizer that is itself frequently near-censored destabilizes the resulting ratio in exactly the settings — small or rural sewersheds — where a stable signal is most needed. A ratio, however constructed, does not automatically recover the source state: it does so only when target and normalizer share the relevant distortion mechanism while retaining sufficiently independent information about the underlying signal, a joint condition that must be checked case by case, not assumed by default.

4.4 When normalization fails

Normalization fails through several mechanisms; **three particularly consequential failure modes** are highlighted here, with the complete eight-mode catalog (including irreversible degradation, unknown non-human sources, and structural model misspecification) given in Supplementary Section S6a. **Correlated numerator-denominator error** arises when target and normalizer are co-sampled from the same physical aliquot or share upstream drivers — a single rainfall event diluting both

simultaneously — so their errors do not cancel independently the way a naive ratio calculation assumes. An **unstable denominator**, one frequently near its own detection limit, inflates ratio variance disproportionately, as noted above. **Event-dependent correction failure** occurs when a target is lost to combined-sewer overflow, or resuspended after a prior sedimentation event: no normalizer, however well matched under ordinary conditions, corrects for mass that never reached the sampling point, or for a signal that reflects historical rather than concurrent source material.

These are not failures of a poorly chosen normalizer; they are conditions under which normalization, as a class of correction, has no purchase at all, and the correct response is flagging the affected period explicitly rather than searching for a more sophisticated ratio. This is the boundary this review insists on throughout: **normalization is a mechanism-specific correction, not reconstruction of the source state**. It can only remove distortion that is actually present in the auxiliary data used, mechanistically linked to the target’s distortion in a way that has been checked rather than assumed, and not itself so noisy or biased that it adds more uncertainty than it removes. Where any of these conditions fails, a normalized signal is not a better estimate of the source state than the raw report was — it is a differently distorted one, and, if reported with unwarranted confidence, a more misleading one.

Table 2. Normalization strategy, corrected mechanism, required assumption and residual uncertainty — synthesized by this review

Strategy	Corrects	Required assumption	Residual/introduced uncertainty
Flow-based load conversion	Dilution volume	Flow measurement is accurate and time-matched to the concentration sample	Flow-sensor error propagates directly into the load; unreliable under high inflow/infiltration or combined-sewer overflow
Census population	Population size (static)	Contributing population is genuinely stable over the analysis period	Silently reinterprets real population fluctuation as a change in source signal
Dynamic population (mobile-device or chemical proxy)	Population size (time-varying)	The proxy tracks the true contributing population in real time	Adds the proxy’s own measurement error; re-identification risk at small spatial scale
Chemical or microbial biomarker ratio	Population size and/or dilution, jointly with the target	Target and normalizer share persistence, partitioning, and temperature sensitivity (fate matching)	Ratio variance amplified near the normalizer’s detection limit; fails outright when fate is mismatched

Performance, cross-scenario, and mechanism-direction statements in the “Residual/introduced uncertainty” column (e.g., “unreliable under high inflow/infiltration,” “fails outright when fate is mismatched”) are pending full-text evidence verification.

5. Reconstruction, identifiability and uncertainty

5.1 Inferential targets and reconstruction models

A wastewater time series can be asked to support several genuinely different inferential targets, and the evidentiary bar rises sharply across them: a **trend** (rising, falling, or flat), a **relative change** (by how much, against a baseline), an **anomaly** (has the signal moved outside an expected range), an **absolute load** (how much material is entering the network), and a **prevalence, incidence, consumption, or exposure estimate** (a population-level quantity in real units). Trend and anomaly detection can be supported by a signal that merely covaries with the underlying process; absolute-magnitude and prevalence-style estimation require the model generating the estimate to be correct, not merely correlated with the target. This ordering is not incidental — it recurs throughout this section as the organizing axis against which every method family, identifiability requirement, and validation level below is actually judged.

Reconstruction methods in the literature span a spectrum of mechanistic commitment — direct mass-balance calculation, convolution models linking a shedding or excretion kernel to source timing, empirical regression against a reference series, mechanistic transport-and-decay models, Bayesian or state-space approaches treating the target as a latent variable, and hybrid process-informed models combining mechanistic structure with a statistical component. Which family is appropriate depends on the inferential target above, not on which method is currently favored: a method well suited to trend detection can be inadequate for absolute-magnitude estimation, and no family in the current literature satisfies every target’s evidentiary requirements simultaneously. A deconvolution approach that grounds its shedding kernel in independently collected clinical data, rather than estimating it jointly from the wastewater series itself (Huisman et al., 2022), is this evidence base’s clearest example of a method deliberately designed around state observability rather than fit quality alone — the distinction §5.2 develops formally.

5.2 Identifiability and observability

Six concepts, distinguished here in full for reconstruction specifically, recur throughout this section. **Structural identifiability** asks whether a model’s parameters would be uniquely determined given noise-free, continuous, infinite data — a property of the model equations alone. **Practical identifiability** asks whether those parameters are actually recoverable, with usable precision, from the finite, noisy data at hand; a structurally non-identifiable model cannot be rescued by collecting more data, while a structurally identifiable one may still be practically non-identifiable under a given program’s sampling frequency and noise level. **Parameter uncertainty** is a distinct question again: how precisely a parameter is estimated once identifiability already holds. **State observability** asks whether the underlying latent trajectory of interest — true prevalence, true consumption — can be recovered from the observed signal at all, a question that remains meaningful even where every parameter is already known; a model can have every parameter pinned down and still have an unobservable state, and conversely a state can in principle be recoverable while individual parameters governing it remain confounded. **Equifinality** names the practical consequence: mechanistically different parameter-and-state combinations can reproduce the same observed data equally well, making a good fit uninformative about which is correct. **Informative data design** — deliberately choosing sampling frequency, auxiliary covariates, or interventions to break these ambiguities — is the constructive response.

The central judgment this section insists on, argued fully once in this manuscript: **a model may fit the observations while the target state remains non-identifiable**. A narrow posterior credible interval or a low residual error is not, by itself, evidence that identifiability has been achieved — it can result from an informative or overconfident prior, an overly rigid model structure, or simply too little independent information in the data to reveal the ambiguity that is actually present. A Bayesian functional-data framework applied to weekly, sometimes-missing wastewater measurements illustrates the point directly: it reports posterior state estimates addressing missingness explicitly, but no formal observability analysis or synthetic-state recovery test accompanies those estimates, so the posterior’s narrowness alone should not be read as demonstrated observability (Dai et al., 2024).

5.3 Validation according to inferential purpose

“Validation” covers at least six distinct exercises in the applied literature, kept separate here. **Internal fit** — goodness of fit on the data used to fit the model — is the weakest, trivially inflatable by adding parameters, and establishes only that the model can reproduce what it was shown. **Temporal holdout** tests generalization across time at the same site, not across setting. **External site validation** — performance at a genuinely different site or population — is the level at which transferability claims are actually tested; a single-site fit, however good, does not establish that a model or threshold transfers elsewhere. **Event-based validation** checks performance specifically around known perturbations (a documented outbreak, a rainfall event) rather than only on routine data. **Synthetic-state recovery** uses a known, simulated ground truth to test whether a method recovers the true state under controlled conditions — the only level that speaks directly to identifiability and observability rather than only to predictive accuracy. **Decision-value validation** asks the furthest-reaching question: whether using the reconstructed signal actually improved a real decision outcome.

Two points recur from the evidence base and are retained here. Predictive performance is not the same evidence as source-state recovery: a machine-learning approach achieving useful short-horizon predictive performance across several sewersheds, evaluated on a genuine train/test split, still contains no individually interpretable parameter and no explicit latent-state representation at all — a legitimate design choice for a forecasting tool, but one that disqualifies it from supporting a mechanistic reconstruction claim (Ai et al., 2022). And the validation level actually reported should match the inferential target claimed, not imply a higher one was met.

5.4 Uncertainty propagation

Uncertainty in a reconstructed quantity is properly a full distribution over the target conditional on every uncertain input and the chosen model structure, not a point estimate with an appended error bar. Propagating it requires accounting for several distinct sources together, not one at a time: **input uncertainty** in every measured or assumed parameter; **correlated parameters**, since inputs sharing an upstream driver do not behave as independent error sources, and treating them as if they were produces a systematically, not just imprecisely, wrong interval; **ratio uncertainty**, which grows disproportionately as a normalizing denominator approaches its detection limit; **censoring and limits of detection**, requiring an explicit censored-data treatment rather than a zero or half-detection-limit substitution; **temporal dependence**, since consecutive samples are rarely independent draws; **model structural uncertainty**, concerning whether the chosen model form is correct at all, which no amount of within-model simulation addresses; and **scenario uncertainty**, concerning which of several discrete external conditions actually applies.

Three families of propagation method are available, distinguished here by what problem each solves. The **delta method** provides a fast, closed-form approximation valid only when the underlying function is close to linear and errors are small and roughly symmetric. **Monte Carlo simulation** handles nonlinear functions and asymmetric distributions by resampling, provided input distributions and their correlation structure are specified rather than assumed independent by default — a formal Bayesian consumption-estimation framework represents excretion-fraction uncertainty as an explicit prior distribution and propagates it through Markov chain Monte Carlo, rather than treating it as a fixed value (Jones et al., 2014). **Bayesian and ensemble approaches** additionally address structural uncertainty, by comparing across candidate model structures rather than characterizing sensitivity within one fixed structure. None of the three, run alone, discharges every uncertainty source above simultaneously.

5.5 From uncertainty intervals to defensible inference

A reported interval is itself a claim requiring its own check, not a self-certifying output. **Calibrated coverage** — whether the true value falls within a reported interval the claimed percentage of the time under repeated application — directly tests whether an interval means what it claims, and is rarely reported in the applied literature surveyed here. **Sensitivity analysis** identifies which inputs actually drive output uncertainty, a different question from which inputs the output is most mathematically sensitive to: a highly sensitive but precisely known input can contribute little to total uncertainty, while an insensitive but poorly known one can dominate it. **Value of information** analysis asks which additional measurement would most reduce decision-relevant uncertainty, directing scarce monitoring investment accordingly. **Uncertainty communication** — translating a propagated interval into a decision-relevant probability rather than a bare error bar — is what connects this section to public-health use. **Decision thresholds** should be set from the asymmetric costs of false alarms against missed detections, not a statistical convention chosen independent of consequence. Finally, some uncertainty is genuinely irreducible given current monitoring design, and the correct response is disclosing a wide interval or a qualitative trend statement rather than reporting false precision. This translation step — from a scientifically honest uncertainty distribution to a statement a decision-maker can actually use — is the hinge this review turns on next.

Table 3. Model family, identifiability requirement, dominant uncertainty and minimum validation — synthesized by this review

Model family	Inferential target	Principal identifiability requirement	Dominant uncertainty	Minimum validation	Principal failure mode
Direct mass balance	Absolute load/consumption	Correction-factor components separably justified, not fixed by assumption	Correction-factor (excretion/pharmacokinetic) uncertainty	Mechanistic consistency check against independent data	Confident wrong estimate under a population-mismatched correction factor
Empirical regression	Trend, relative change	Not applicable (no mechanistic parameters)	Reference-data uncertainty	External site or temporal holdout	Overfitting to one site/period, no transfer
Convolution/deconvolution	Relative change, incidence	Kernel and state jointly (not separably) estimable without external grounding	Kernel/state equifinality	Synthetic-state recovery or externally grounded kernel	Kernel-state trade-off masked as one precise fit
Mechanistic transport-decay	Absolute load	Structural adequacy of the process representation	Structural/model-form uncertainty	Event-based validation	Confidently wrong output under an unmodeled process
Bayesian/state-space	Prevalence, incidence	Practical identifiability under the chosen prior	Narrow-posterior-as-false-certainty risk	Posterior predictive check, prior sensitivity analysis	Overconfident interval from an informative prior

Model family	Inferential target	Principal identifiability requirement	Dominant uncertainty	Minimum validation	Principal failure mode
Hybrid process-informed	Absolute load, prevalence	Cross-site generalization of the learned component	Temporal leakage; structural uncertainty of the learned part	External site and decision-value validation	Inflated apparent skill from temporal leakage

Every “Dominant uncertainty,” “Minimum validation,” and “Principal failure mode” entry in this table is this review’s own proposed classification and is pending full-text evidence verification against each model family’s underlying studies. No cell should be read as an independently confirmed finding.

6. From wastewater signals to public-health action

6.1 Surveillance objectives and decision thresholds

Trend monitoring, anomaly detection, and absolute-magnitude estimation are three distinct surveillance objectives, each tolerating a different uncertainty structure and requiring a different threshold-design logic; a program should name which objective a given signal serves before setting a threshold, not after — anomaly detection in particular is a formal statistical-process-control problem, and a detection rule tuned on a symmetric-noise assumption misbehaves against the right-skewed, detection-limit-censored data this evidence base repeatedly documents as the norm (Assoum et al., 2023). A detection threshold should be set from the asymmetric costs of a false alarm against a missed detection (§5.5), not an arbitrary statistical convention. A sensitivity/specificity pair calibrated at one site is a property of that joint system — signal, noise structure, and threshold together — not a universal property of “wastewater-based detection”: a recent preprint reports a sensitivity advantage for an event-detection framework over a clinical-data baseline across a large U.S. county sample (Link, Garrido, et al., 2026), but this is a single, non-peer-reviewed study set in one national context, and **threshold transportability across populations and sewersheds cannot be assumed from it and must be re-validated in each new deployment context.**

6.2 Multi-source interpretation

No surveillance system treats a single stream as sufficient. Clinical case and testing data, hospital admissions and pharmacy sales, mobility data, weather, and demographic context each contribute a distinct, partially independent proxy for the same underlying process, and none is a ground truth against which wastewater should be one-sidedly validated — each carries its own bias and lag. A four-system integration of hospital, wastewater, meteorological, and internet-search data into one shared time series found a substantial subset of the fused variables correlating significantly with clinical case counts (Zhang et al., 2025), illustrating that fusion can outperform any single stream. But **fusing these streams does not automatically remove shared confounding, differential time lag, or selection bias.** A shared external driver — a policy change, a media event — can move several streams together for reasons unrelated to the true signal, producing the appearance of multi-source corroboration where none exists; and streams that lead or lag each other by an uncertain, setting-dependent interval will be systematically misaligned by any fusion approach that assumes a single fixed lag. Weather deserves particular care in this respect: it is not merely another corroborating stream but, per Part 3, a direct confounder of the wastewater signal itself, so a fusion pipeline that treats a rainfall-driven dip as independent “corroborating” evidence rather than as amplitude distortion needing correction risks double-counting the same event twice.

6.3 Actionability, limits and ethics

A signal can be scientifically valid and still not be actionable. Four conditions govern the gap: the uncertainty has been translated into decision-relevant terms, not left as a bare point estimate; a named institutional actor and response protocol exists to receive the signal; the detection-to-action interval is shorter than the timescale over which the underlying condition changes; and the alert has been validated against the cost structure of its specific deployment, not borrowed unmodified from elsewhere.

Wastewater surveillance cannot, by construction, provide an individual diagnosis or an individual care decision — it is a population-level, aggregated signal, and that aggregation is the same property that makes its no-consent advantage possible. That advantage is unevenly available: non-sewered populations are structurally, not merely incidentally, invisible to this method, and a national assessment finds sewer connectivity systematically, not randomly, associated with specific demographic and geographic factors (Yu et al., 2024). As spatial resolution narrows toward a single building, three distinct problems emerge that need different mitigations, not one undifferentiated “privacy” response: individual **re-identification risk** — a documented instance exists of a wastewater detection being traced upstream to a specific residence (Moallef et al., 2025) — **group-level stigmatization** independent of any individual being identified, where a defined population is portrayed as “diseased” regardless of whether anyone in it is actually re-identifiable, and **structural equity** in which communities are monitored at fine resolution at all, since deploying building-scale monitoring unevenly redistributes both its surveillance benefit and its privacy risk. A decision-ready signal, at minimum, should disclose its surveillance objective, threshold and calibration basis, fusion status, spatial resolution, and the named actor and protocol that acts on it.

7. Enabling infrastructure and technology readiness

7.1 Process-resolving infrastructure

Distributed, upstream sampling disaggregates a single citywide signal into smaller, traceable sub-catchments, trading spatial resolution against signal stability — finer granularity means less averaging-out of the aleatory variability Part 2 already established, not less uncertainty overall. A passive-sampler validation across nested lot, suburb, and city scales demonstrates early detection capability specifically at the fine spatial scales composite autosamplers deploy poorly (Schang et al., 2021), directly illustrating this trade-off. Online covariates and near-real-time measurement compress the latency between sample collection and usable output, which matters directly for the detection-to-action timescale named in §6.3, but a faster wrong number is not an improvement over a slower correct one. Digital twins — continuously updated hydraulic models of the sewer network, an active and rapidly growing research area (Bam et al., 2025) — supply the residence-time and scenario information Part 3’s process-informed correction argument requires, functioning as a process-constraint-generating tool rather than an automatic validator of the true source signal. Event-responsive monitoring, adjusting sampling around storm events rather than holding a fixed schedule, directly targets the scenario uncertainty named in §5.4.

7.2 Process-informed analytics

Hybrid mechanistic-statistical models and physics-informed machine learning incorporate transport and decay structure as a constraint on, rather than a replacement for, a learned component (Pagsuyoin et al., 2025); the same cross-site and cross-population transfer requirement §6.1 sets for detection thresholds applies to any learned component without modification, and a temporal-leakage check — confirming a model was not trained on information unavailable at true prediction time — is a

minimum requirement before any performance claim is credited. Interoperable metadata, standard reference materials, and interlaboratory comparability are infrastructural, not analytical, prerequisites: a national metrology working group argues standardized reporting and proficiency testing are necessary, not merely desirable, preconditions for cross-laboratory data to be used reliably in public-health decisions (Keenum et al., 2024), converting a single laboratory's internal precision into cross-study comparability, without which cross-site validation itself cannot be performed with confidence.

7.3 Readiness and equitable deployment

A single favorable result at one level of maturity is evidence for that level only, not for the levels above it. This review adapts a readiness hierarchy — **analytical detection** → **field validation** → **cross-site validation** → **operational integration** → **demonstrated decision value** — and insists throughout: **analytical novelty is not equivalent to epidemiological validity, operational readiness, or demonstrated decision value**. A method can be highly novel analytically while remaining entirely unvalidated at the operational or decision-value level, and no technology surveyed in this review's evidence base is confidently placed above the middle of this hierarchy for routine public-health use. Equitable deployment is itself a readiness dimension, not a separate afterthought: infrastructure investment concentrated in already-well-resourced sewersheds reproduces the same structural-equity problem named in §6.3, regardless of how analytically mature the underlying technology is.

8. Research priorities and conclusions

8.1 Testable research priorities

1. **Paired source-sewer-outlet observations.** Unresolved: how much amplitude/temporal distortion accumulates at each named stage individually. Design: simultaneous sampling at excretion-adjacent, mid-network, and outlet points on matched events. Gain: direct, stage-attributed distortion estimates rather than inferred ones.
2. **Rainfall-event mass-balance experiments.** Unresolved: whether process-informed rainfall correction generalizes beyond one calibration program. Design: paired dry/wet-weather mass-balance trials across contrasting network types. Gain: a transferable, tested correction model rather than a single-program result.
3. **Target-normalizer fate-matching studies.** Unresolved: which target-normalizer pairs genuinely share persistence and partitioning. Design: co-spiked decay and partitioning trials across matrices. Gain: an evidence-based, not assumed, fate-matching criterion for normalizer selection.
4. **Synthetic-state recovery and observability tests.** Unresolved: whether current reconstruction methods actually recover a known state. Design: apply candidate models to simulated data with a known ground truth. Gain: direct identifiability/observability evidence, not inferred from fit quality alone.
5. **Cross-city and cross-climate external validation.** Unresolved: whether thresholds, machine-learning and hybrid process-informed models, and correction models transfer across settings. Design: pre-registered multi-site validation using one model, multiple independent sites. Gain: a tested, not assumed, transportability finding.
6. **Uncertainty-coverage and decision-value trials.** Unresolved: whether reported intervals achieve their claimed coverage and change real decisions. Design: prospective coverage checks paired with a decision-outcome comparison against a non-wastewater-informed baseline. Gain: evidence connecting propagated uncertainty to actual decision value, not just to a wider interval.

8.2 Conclusions

This review reframes wastewater-based epidemiology as an inverse problem, in which a source-end signal is recoverable from a downstream measurement only to the extent the intervening chain of excretion, transport, sampling, and analysis remains identifiable. Four implications follow. First, the amplitude-temporal-spatial distortion taxonomy developed here gives every stage of that chain a shared vocabulary, replacing a stage-by-stage list of disconnected error sources with one consistent framework applicable from source-end shedding heterogeneity through sewer transport to analytical measurement. Second, normalization should be understood as a mechanism-specific causal correction, not a reconstruction of the source state; it removes only the distortion it is mechanistically linked to, and applying it as a general-purpose fix — assuming any plausible auxiliary variable will do — is a documented, recurring source of error across the normalizer comparisons this review surveys. Third, reconstruction claims require distinguishing identifiability, observability, and validation level from mere predictive performance or a narrow uncertainty interval, since none of the latter, by itself, establishes the former, and a model may fit the data well while the state it claims to recover remains genuinely unresolved. Fourth, translating an uncertainty-quantified signal into public-health action requires an explicit account of actionability, transferability, and equity, not only of statistical confidence — a scientifically valid signal is not automatically a decision-ready one.

Taken together, these four propositions target one recurring failure mode in the applied literature this review surveys: a result established at one stage of the source-to-decision chain — a detection, a correlation, a narrow interval — read as though it settled a question belonging to a later, evidentially distinct stage. Closing that gap, stage by stage, rather than assuming later stages inherit earlier ones' confidence, is this review's central contribution. The research priorities above name where the evidence to close that gap most urgently needs to be generated, and until it is, every quantitative claim in this synthesis should be read as this review's own conceptual organization of an evidence base that has not yet been formally, systematically verified.